

Action plan to prevent the effects of heat waves on health (POCS)

Summer 2022

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Federation of Municipalities of Catalonia (FMC)
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Institute of Legal Medicine and Forensic Sciences of Catalonia (IMLCFC)
Health and Social Consortium of Catalonia (CSC)
Catalan Union of Hospitals
Barcelona Social Services Consortium (CSSBcn)
Red Cross
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[Action plan to prevent the effects of heat waves on health \(POCS\)](#)

Summary

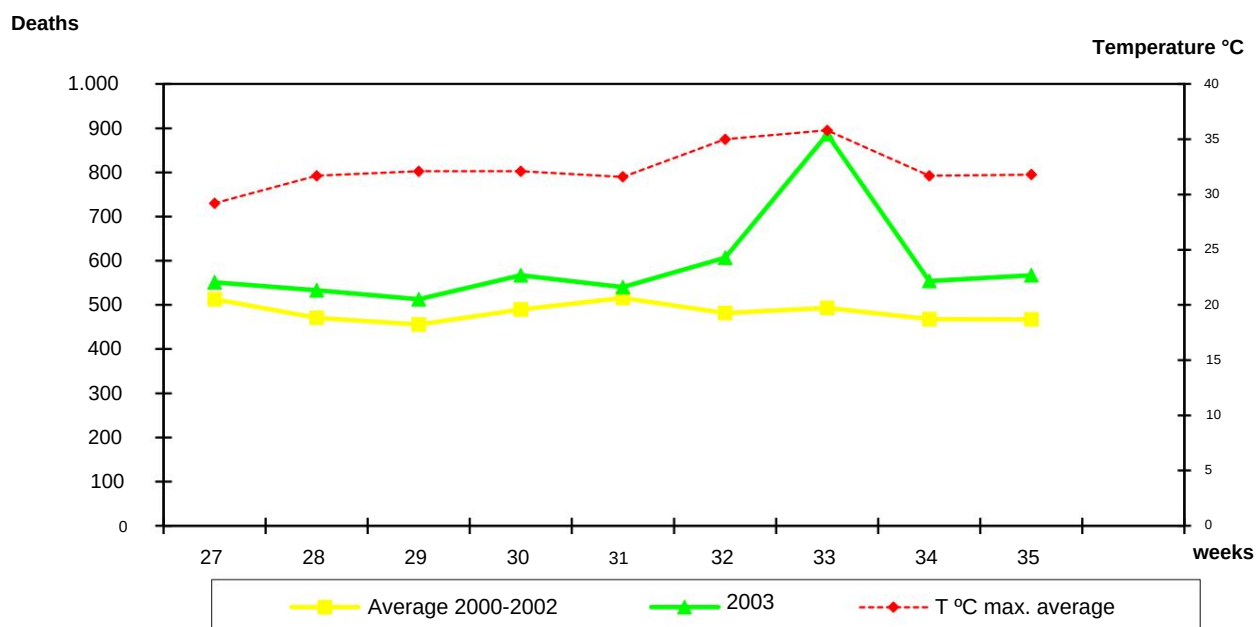
1. Antecedents	5
2. Objectives	6
3. Actions to carry out and achieve the objectives set	7
Objective 1	7
Objective 2	10
Objective 3	13
4. Action phases	14
Phase 0	14
Phase 1	16
Phase 2	17
5. Bibliography	20
6. Links of interest	23
7. ANNEXES	24
Annex 1 . Recommendations for the population	24
Annex 2 . Posters and video for summer 2022	27
Annex 3 . Leaflet from the General Directorate of Social Protection of the Department of Labor, Social Affairs and Families	29
Annex 4 . Documents of the Catalan Institute for Occupational Safety and Health Department of Labor, Social Affairs and Families	30
Annex 5 . Risk groups and recommendations for professionals	32
Annex 6 . Heatstroke notification form	39
Annex 7 . Heatstroke Death Notification Form (IMLCFC)	41

1. Antecedents

For years, the negative effects of thermal variations on health. Excessive and sustained heat leads to increased mortality, especially in the elderly and people with chronic diseases. It is known that high and sustained heat for a few days causes excess mortality which varies between 12% and 40% in developed countries.

The high temperatures experienced in Europe in the summer of 2003 had a effect on the increase in mortality in different countries, including ours. In Catalonia, during the weeks of August 3 to 16, when maximum temperatures were reaching 35° C on average, mortality values were exceeded by 53% compared to those registered during the same weeks in 2002.

Figure 1. Deaths to residents of seven municipalities of Catalonia (July-August 2003)



In addition to an excess of mortality, an increase in the number was also observed of unscheduled admissions to hospitals in Catalonia, which increased by 7.5% in July and August 2003 compared to the same months in 2002. The increase of income was concentrated during the period between August 7 and 20, coinciding with the highest and most sustained thermal increases.

With this background, on May 26, 2004, it was launched for the first time Action Plan to prevent the effects of heat waves on health (POCS) which was carried out again in the following years.

The actions resulting from the implementation of this Plan during the summer of 2022, are summarized in the document "Report on the actions carried out and the results obtained. Summer 2022" which you will find on the website of the Public Health Agency of Catalonia (ASPCAT) >Public Health Surveillance>Other programs> [POCS](#).

2. Objectives

The objectives of the Action Plan to prevent the effects of heat waves on health (POCS) are these:

1. Predict possible events as far in advance as technical means allow.
dangerous weather situations.
2. Minimize the negative effects of heat waves on public health
population of Catalonia, especially the most vulnerable groups: the elderly (especially those over 75), babies, people with disabilities
physical or mental disabilities who have limited self-care, people with chronic diseases, frail people living alone, people with conditions precarious social situations, patients with medications that act on the system central nervous system, people who practice intense physical activity or those who must remain or do physical activity outdoors (including work activities), among others.

3. Coordinate the measures and resources available in Catalonia to deal with the possible heat wave.

3. Actions to carry out and achieve the objectives set

Objective 1

Predict as far in advance as technical means allow possible weather situations of heat hazard (from June 1 to June 30) September).

In order to successfully carry out this objective, it is necessary to carry out the following actions: following:

1. Collect the observed daily temperatures (maximum and minimum) and humidity, as well as the forecasts 48 hours in advance for ten cities in Catalonia: Barcelona, Hospitalet de Llobregat, Manresa, Mataro, Santa Coloma de Gramenet, Sabadell, Terrassa, Girona, Lleida and Tarragona.
2. Evaluate forecasts based on **advance notice** (between 36 and 96 hours) of anticipation) and **warnings** (between 0 and 60 hours of anticipation) of **situation meteorological hazard (SMP)**.

SMPs can have **six degrees** of probability of occurrence.

Grades **1 to 3 of SMP** indicate different **probabilities of occurrence of extreme maximum temperature**.

Level 1 (in yellow): probability of extreme maximum temperature lower than and 30%,

Level 2 (in yellow): probability of extreme maximum temperature between 30% and 70%, and

Action plan to prevent the effects of heat waves on health (POCS)
Department of Health. Government of Catalonia

Level 3 (in orange): probability of higher extreme maximum temperature to 70%.

When there is a grade 1, 2 or 3, it corresponds to a **warning of a low threshold SMP**.

Grades **4 to 6 of an SMP** indicate different **probabilities of occurrence of a heat wave** (three consecutive days of extreme maximum temperature).

Level 4 (in orange): probability of heat wave less than 30%,

Grade 5 (in red): probability between 30 and 70%, and

Level 6 (in red): probability of heat wave greater than 70%.

When there is a grade 4, 5 or 6 it will correspond to a **warning of a high threshold SMP**.

Figure 2. Degrees of danger according to probability of occurrence

 SENSE PERILL (0)	 PERILL ALT (3-4)
 PERILL MODERAT (1-2)	 PERILL MOLT ALT (5-6)

LLINDAR				
llindat alt	4	5	6	
llindat baix	1	2	3	
SENSE AVÍS	0			
	BAIXA (10-30%)	MITJANA (30-70%)	ALTA (> 70%)	PROBABILITAT

Accordingly, the possible situations are:

- a) Extreme maximum temperature warning (threshold SMP warning low):** this warning must be activated when in the following 60 hours it is

expected to exceed the **98th percentile of the maximum daily temperature of the summer period.**

b) Heat wave warning (high threshold SMP warning): must be activated this notice when the overcoming is expected in the following 60 hours of the 98th percentile of the maximum daily temperature of the summer period during **three consecutive days.**

SMP notifications can be enabled or disabled independently of each region.

It should be noted that, when an SMP notice groups a set of counties with the same probability of exceeding a certain meteorological threshold, it means that it is expected that within the area formed by this group of counties it can the threshold is exceeded, at some points, with the assigned probability, but not that must happen in all the counties that make up it.

3. Perform **daily monitoring of the funeral data** of the same ten municipalities, including deaths, age and sex.
4. Additionally, **activity and mortality data** are available.
daily hospital care in four hospitals in Barcelona.
5. Conduct **daily monitoring of judicial deaths due to heatstroke** throughout Catalonia. From the Institute of Legal Medicine and Forensic Sciences of Catalonia (IMLCFC) reports deaths from heatstroke in the Sub-Directorate of Surveillance and Response to Public Health Emergencies (SGVRESP).
6. Monitor **heatstroke** and its evolution, within the framework of Epidemiological Surveillance Network of Catalonia (XVEC). Hospitals report heatstroke, and its evolution, to the surveillance services corresponding epidemiological (SVE), which notify them to the SGVRESP.

7. To be coordinated with the *"Ministry of Health, Consumer Affairs and Social Welfare"*, within the framework of the National Plan of preventive actions against the effects of the effects of excessive temperatures on health.

Objective 2

Minimize the negative effects of heat waves on public health population of Catalonia.

In order to successfully carry out this objective, it is necessary to carry out the following actions: following:

1. Dissemination of recommendations to the population to prevent the effects of heat about health

a) **Recommendations** for the general population from the **website of the Agency for Public Health of Catalonia (ASPCAT)** >Public Health Surveillance>Others programs> FEW ([annex 1](#)).

b) Distribution of the poster ["Health and enjoy the summer"](#) in Catalan, Aranese and Spanish ([annex 2](#)) on the website of [the Public Health Agency of Catalonia \(ASPCAT\)](#) so that they can be used in **healthcare centers**, offices **pharmacy**, among others.

A **video with advice** is made available to health centers basic for those who have screens in waiting rooms. This material can be found on the ASPCAT website and on [video health](#). At the offices of pharmacy, it is necessary to provide information to the population. Likewise, colleges of pharmacists will influence recommendations to minimize the effects negatives of the heat wave, so that pharmacy offices can continue to adequately inform the population.

c) **Video capsule** that ASPCAT makes available to healthcare centers, town halls, YouTube, among others. The video capsule is 1.5 minutes long

and can be broadcast in different public places: subway screens, airport, among others others. It can be found on the video [health website](#).

d) Information with recommendations is provided through the **061 CatSalut** service **Answer**. This service will answer all the queries of the population related to a possible heat wave.

e) Recommendations for all assisted services for the elderly and disabled, owned and managed by the **General Directorate of Social Protection from the Department of Labor, Social Affairs and Families**.

The **leaflet**, "[Environmental action measures to prevent heat](#)", in all centers, through their website (annex 3).

f) Recommendations in the field of work. **The Catalan Institute for Safety and Health Labor** of the **Department of Labor, Social Affairs and Families**, disseminates its recommendations leaflet "[Work and the heat: more prevention in summer](#)" and from the infographic "[Protect your skin from the sun's ultraviolet radiation](#)" (Annex 4), and disseminates information about the working day and recommendations in the summer on the website.

- Improvement and updating of the ICSSL website on "[Heat during the day summer job](#)" which includes main recommendations that must be taken into account It takes into account both businessmen and workers.

The heat during the working day in the summer [See the information](#).

Action plan to prevent the effects of heat waves on health (POCS)
Department of Health. Government of Catalonia

-Dissemination of the signature and updating of the website with the inclusion of all documents and recommendations from the ICSSL communication mailbox.



-Dissemination, through the collaboration of mutual collaborators with the Social Security (MCSS) to companies that have associates from documents, recommendations and links to prevent the effects of heat during the working day. Once the summer period is over, these entities they report to us what dissemination they have done by email to companies according to size and territorial distribution.

-Analysis of work accidents registered in Catalonia linked to the heat, in the hot period.

-Publication of tweets related to the heat wave campaign, mainly coinciding with heat warnings.

2. Recommendations for health care personnel working with at-risk populations in primary care centers (CAP), mental health centers (CSM), centers social health services (CSS), and hospitals (Annex 5). The **definitions of heat stroke and other disorders caused by heat.**

3. Updating the censuses of fragile people in town halls, centers
primary care and social services in order to intervene.

Objective 3

Coordinate the measures and resources available in Catalonia to deal with the possible heat wave.

- The coordination of the POCS is carried out by the Public Health Agency of Catalonia and the Catalan Health Service. The following organizations and departments are coordinated: Meteorological Service of Catalonia, Department of the Interior, National Plan of emergencies of Catalonia (PLANUC), 061 CatSalut Respon i Emergencies Medical, Department of Labor, Social Affairs and Families, personal services of Barcelona, Catalan Association of Municipalities, Federation of Municipalities of Catalonia, Council of Pharmaceutical Colleges of Catalonia, Institute of Medicine Legal and Forensic Sciences of Catalonia, Health and Social Consortium of Catalonia, Barcelona Social Services Consortium, public health services of the Provincial Council of Barcelona, Catalan Union of Hospitals, Social Services Consortium of Barcelona, Red Cross and the Epidemiological Surveillance Network of Catalonia.

The activities carried out to achieve the coordination of measures to deal with possible heat waves are fundamentally the following:

- Review of the Plan and finalization.
- Meeting prior to the start of the POCS application season to implement surveillance, communication and alert mechanisms necessary within the Plan. This meeting also evaluates the season anterior.
- Meetings, if necessary, between the different organizations and participating departments in order to carry out objectives 1 and 2, and especially in case a heat wave occurs.

4. Action phases

The action phases are distributed in an indicative manner between start and end dates. completion but dates could change depending on observed temperatures

Phase 0

This phase will run from **May 1st until June 15th, and from August 31 to September 30**. It is considered the preparatory and basic phase of the plan.

It includes the following actions:

- a) Coordination meetings between those involved.
- b) Update of the Action Plan to prevent the effects of heat waves on health. **Dissemination** to health centers.
- c) **Adaptation** in each health center: CAP, mental health centers (CSM), outpatient clinics, PADES teams, acute care centers, social and health centers inpatient care, inpatient CSM and geriatric residences of a **plan of action** with measures that guarantee sufficient support for people large and fragile. The Plan must include:
 - a. General preventive measures
 - b. Specific ones for people at risk (Annex 5)
 - c. The communication circuit with social services (cross-referencing of censuses) i
 - d. The actions planned in the event of a heat wave.
- d) **Preparation** in each CAP of an **updated census of the people with the most a risk** that must be contrasted with the social services of the territory.

Action plan to prevent the effects of heat waves on health (POCS)
Department of Health. Government of Catalonia

- e) The Department of the Interior has a **tool on its website for updating from the census to the municipalities.**
- f) The Department of the Interior has at the disposal of the municipalities the ***Guide for the municipal action procedure for heat waves***, which was elaborated during the years 2009 and 2010. This guide places special emphasis on the municipal organization for emergency management so that the response municipal is planned and sized correctly.
- g) Sending of **information letters to the town councils** by the Association Catalan Association of Municipalities and the Federation of Municipalities of Catalonia.

From June 1st:

- a) **Communication to the general population.** Dissemination of recommendations to the population to prevent the effects of heat on health.
- b) Daily collection of **temperature data** provided by the Service Meteorological Department of Catalonia and **deaths** (data obtained from the funeral homes) in the ten municipalities subject to control.
- c) Collection of **emergency care activity and hospital mortality** in four sentinel hospitals.
- d) Collection of **heatstroke diagnosed and** reported within the framework of the XVEC up to the SGVRESP. **The data collection sheet is in Appendix 6.**
- e) Collection of judicial deaths from the same ten municipalities and deaths by **heatstroke** throughout Catalonia. The **data collection form** is located at **l'annex 7.**

Phase 1

This **phase will be active** during the period **from June 15 to August 31**, and foresees the following actions:

- a) Weekly disclosure of the qualitative assessment of temperature data and deaths.
- b) Dissemination of preventive recommendations for the general population, and special attention to fragile people at high social risk.
- c) Implementation of general and specific measures by the services healthcare, both outpatient and home, in order to reduce the risk of heat excessive pressure on very vulnerable people. These measures have been to be applied to all CAPs, acute hospitals, socio-health centers, CSM and geriatric residences.
- d) In pharmacy offices, information and advice is provided to the population who request it. sol·liciti.
- e) The Health and Social Consortium of Catalonia disseminates information about the POCS (Plan documentation, campaign material and web link) to its associated centers (Health and dependency care centers) in through the usual dissemination channels: email, CSC Newsletter, website, Twitter, from the moment the campaign is activated.
- f) The Catalan Hospitals Union disseminates information and monitors the POCS to the health and social sector centers that are associated with it.
- g) The public health services of the Barcelona Provincial Council disseminate information to their web space, in its virtual community and in the Provincial Council's Intradiba.
- h) The Catalan Association of Municipalities informs all town halls about the POCS and associated local entities via email and links to the website.

Action plan to prevent the effects of heat waves on health (POCS)
Department of Health. Government of Catalonia

i) The Federation of Municipalities of Catalonia disseminates information on the POCS through the website.

j) **061 CatSalut Respon** participates in the POCS with three well-defined lines of action defined:

- a. Information to citizens about the effects of a possible wave of heat can cause on health and tips to prevent these effects.
- b. Information on the measures to take in the event of heatstroke.
- c. Attention to consultations received for heat-related disorders, either by telephone resolution with a health council, or by activating some resource (medical personnel at home or ambulance from any type) in case transfer to a hospital is necessary.

k) **Red Cross**: gives prevention talks to the general public where they provide advice for prevent the effects of heat, make telephone follow-up calls to fragile people, home visits among the most fragile group, etc.

Phase 2

This phase **should** only be activated **in the event of an alert**, when a rise in temperatures: **low threshold SMP and high threshold SMP**.

When faced with a **low-threshold SMP**, one must be aware of the situation and be alert in case the situation persists. The actions that are carried out are the following following:

- a) Notification to healthcare facilities that there is an alert so that they are prepared action plans in case the dangerous situation continues.

Action plan to prevent the effects of heat waves on health (POCS)
Department of Health. Government of Catalonia

b) **061 CatSalut Respon** increases in this phase its activity already described in phase 1.

c) **Red Cross**. Increases its activity through telephone advice.

d) The **dissemination to the population of the advice** to follow in the event of temperatures extremes.

In the event of a high threshold SMP (heat wave) the following actions are carried out:
following:

a) Activation of specific action plans planned in the event of a heat wave in every health center, internment centers and geriatric residences.

b) Specific active interventions in the field of **primary health care**, both at home and outpatient, for the most **fragile** people according to the census.

c) Intervention of **social services** in the detection of the most **fragile** people by in order to give them the most appropriate attention.

d) Redistribution, in detention and residential centers, of the most vulnerable people fragile towards air-conditioned areas.

e) **061 CatSalut Respon** increases in this phase its activity already described in phase 1.

f) **Red Cross**. Increases its activity, through telephone advice and calls monitoring of vulnerable people.

g) **Notice to the media** .

In the event of a high-threshold SMP, emergency resources are activated: the Emergency Response Center. Operational Coordination of Catalonia (CECAT) of the Department of the Interior upon receiving information from the Public Health Agency of Catalonia activates the Civil Protection Plan of Catalonia (PROCICAT), medical emergencies and the Red Cross. In this case, it is would take the relevant measures simultaneously and in a coordinated manner, both for

Action plan to prevent the effects of heat waves on health (POCS) Department of Health. Generalitat de Catalunya

regarding out-of-hospital care and emergencies (through emergencies medical and Red Cross), as well as in healthcare centers, through the territorial network of the PLANUC (the health regions of the Catalan Health Service).

Action plan to prevent the effects of heat waves on health (POCS)
Department of Health. Government of Catalonia

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Action plan to prevent the effects of heat waves on health (POCS)
Department of Health. Government of Catalonia

6. Links of interest

- [FEW](#)
- [Cheers and enjoy the summer](#)
- [Ministry of Health, Consumption and Social Welfare](#)
- [WHO \(PUBLIC HEALTH ADVICE\)](#)
- [WHO \(Heatwaves and Health: Guidance on Warning-System Development\)](#)
- [Health video](#)

7. ANNEXES

Annex 1. Recommendations for the population

A healthy summer

Heat waves: a health hazard

Very extreme temperatures and excess heat cause a loss of fluids and mineral salts (chlorine, potassium, sodium, among others) necessary for the body.

This can aggravate a chronic illness or cause dehydration and exhaustion.

If exposure to such high temperatures is prolonged, you can suffer heat stroke, which is a situation that can end up being serious.

There are some symptoms that can alert us that someone is suffering from heatstroke: very high temperature, headache, nausea, intense thirst, convulsions, drowsiness or loss of consciousness. If you see these symptoms, move the person to a cooler place, give them water, wet them and air them out, and **urgently notify the health services (061)**.

Pay attention to the following tips, especially if you are among these groups or are you family or acquaintances of these people:

1. People over 75 years of age.
2. People with certain social circumstances (living alone, poverty, etc.) others).
3. People with physical or mental disabilities and mobility limitations or self-care.
4. Babies.
5. People with high blood pressure, diabetes, cardiovascular diseases, respiratory diseases (COPD, bronchitis), kidney diseases, Parkinson's, Alzheimer's, obesity or other chronic diseases.
6. People who take medication that acts on the central nervous system (tranquilizers, antidepressants, psychotropics) or diuretics.
7. People with mental and behavioral problems due to the use of psychoactive substances or alcohol.
8. People who engage in intense physical activity.
9. People who must remain or practice physical activity outdoors (including work activities).

Action plan to prevent the effects of heat waves on health (POCS)
Department of Health. Government of Catalonia

Tips to avoid problems caused by heat

Protect yourself from the sun and heat.

At home, control the temperature:

- During sunny hours, close the blinds on windows where it is hot.
- Open the windows of your house at night to cool it down.
- Stay in the coolest rooms.
- Use some type of air conditioning to cool yourself and the environment (fans, air conditioning, fans, etc.). If you don't have air conditioning, try to spend the hottest hours in air-conditioned places (libraries, shopping centers, cinemas, among others).
- Refresh yourself often with showers, wet towels, etc. others.

On the street, avoid direct sunlight:

- Wear a cap or hat.
- Wear light clothing (such as cotton), light colors, and that is not adjusted.
- Try to walk in the shade, stay under an umbrella when you are on the beach and rest in cool places on the street or in closed spaces that are air conditioned.
- Bring water and drink it often.
- Wet your face and even your clothes a little.
- Be careful when driving during the sunniest hours and do not leave children or fragile people with the windows closed.

Reduce physical activity during the hottest hours.

- Avoid going out during the midday hours, which is when it is hottest.
- Reduce intense activities.

Drink plenty of water and watch your diet.

- Drink water even if you are not thirsty.
- Do not drink alcoholic beverages.
- Avoid very hot foods and those that provide a lot of calories.

Action plan to prevent the effects of heat waves on health (POCS)
Department of Health. Government of Catalonia

Help others.

If you know elderly or sick people who live alone:

- Try to visit them once a day.
- Help them follow these tips.
- If you are taking medication, check with your doctor whether it may influence thermoregulation and whether it needs to be adjusted or changed.

Inform yourself well.

- Find out about the opening hours of the **primary care centres** closest to you, both in your municipality and where you are going on holiday.
- **061** Remember this CatSalut Respon number , which can assist you daily, 24 hours a day. They can give you health advice, **point you to the nearest health centers or, if necessary, activate home care.**
- Remember that at **pharmacies** you can obtain information on how to avoid problems caused by heat.
- Follow [weather forecasts](#) that they can anticipate for you hottest days:
- Go to health and social professionals in advance to prevent possible problems for your health.

For more information:

[Cheers and enjoy the summer!](#)

[Heat stroke](#)

[Heatstroke prevention by the Ministry of Health](#)

Action plan to prevent the effects of heat waves on health (POCS)
Department of Health. Government of Catalonia

Annex 2. Posters and video for summer 2022

Poster: Cheers and enjoy the summer! (Catalan)



Cheers and enjoy the summer! (Aranese)

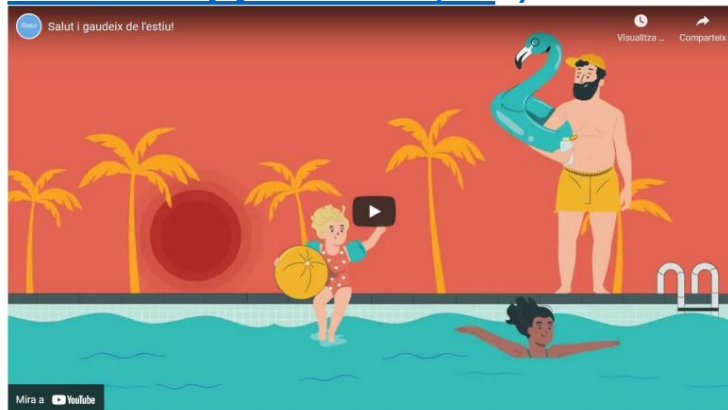


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Cheers and enjoy the summer! (Castilian)



Cheers and enjoy the summer (video)



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Annex 3. Leaflet from the General Directorate of Social Protection of the Department of Work, Social Affairs and Families



MESURES D'ACTUACIÓ AMBIENTAL PER PREVENIR LA CALOR

La hipèrtermia és l'augment anormal de la temperatura corporal a causa de l'exposició a temperatures ambient elevades.

La regulació de la temperatura corporal en la persona gran és una condició indispensable per al bon funcionament integral.

1 FACTORS DE RISC EN LES PERSONES GRANS

- Problemes de circulació perifèrica
- Malalties neurodegeneratives
- Malalties cròniques: EPOC, HTA, cardiovasculars...
- Excés de medicaments: polifarmàcia
- Sobrepès o manca de pes adequat

2 SÍNDROMES CLÍNQUES CAUSADES PER LA CALOR

- Rombes o les canes
- Esgotament
- Mal de cap
- Vòmits i mareig
- Malestar i malalties
- Fèbre

3 NORMES A SEGUIR

En cas de temperatura ambiental alta

- Ingerir líquids, fins i tot sense tenir set
- Evitar menjar calent
- Fer servir vestits lleugers i evitar peces tancaïdes
- Vestir-se amb roba de cotó
- No fer exercici vigorosos
- No exposar-se al sol
- Refrescar-se amb aigua freda i cremes hidratants
- Utilitzar capells i para-sols

A l'habitatge propucarem

- Refrigerar les estances
- Posar cortines i para-sols a les finestres més assolides
- Obrir finestres a la nit per afavorir el corrent d'aire
- Mantenir a les fosques les habitacions amb més radiació diürna
- Evitar l'aglomeració de persones en espais reduïts

En cas d'hipèrtermia

- Avisar els serveis mèdics
- Aplicar els tractaments prescrits
- Col·locar draps humits i freds a diverses parts del cos
- Canviar els bolquers sovint
- Mantenir els llençols del llit secs
- Hidratar la pell i ingerir aigua
- Canviar de postura la persona allitada
- Afavorir la ventilació

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Department of Health. Government of Catalonia

Annex 4. Documents from the Catalan Institute for Occupational Safety and Health of the Department of Labor, Social Affairs and Families

Information document: Work and the heat: more prevention in summer

El treball i la calor: més prevenció a l'estiu

quaders de prevenció

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Generalitat de Catalunya
Departament d'Empreses i Ocupació
Secretaria d'Innovació i Recerca Laboral

Subdirecció General de Seguretat i Salut Laboral
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Torre de Galdic, 303 08028 Barcelona Tel. 93 401 30 00	Alvarez de Castro, 4, 2n 17001 Girona Tel. 972 20 59 33	Riu Boix, 2 25001 Lleida Tel. 973 21 63 80	Vidal i Barroque, 20 43005 Tarragona Tel. 977 23 58 25		

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Departament d'Empreses i Ocupació

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DISENY
IDOC - Estudi Vaguet

amb agraïment a tots els subjectes a una Biblioteca Recerca i Innovació Social i Cultural sense fins. Després de la seva creació, la Biblioteca Recerca i Innovació Social i Cultural sense fins ha estat incorporada a la Biblioteca Recerca i Innovació Social i Cultural sense fins. La Biblioteca Recerca i Innovació Social i Cultural sense fins es pot consultar a <http://www.biblioteca-social.cat>

quaders de prevenció

Infographic: Protect your skin from the sun's ultraviolet radiation

Protegeix-te la pell de la radiació ultraviolada del sol
Excés de sol = cremades = càncer de pell

**Generalitat de Catalunya
Departament de Treball,
Afers Socials i Famílies**

a la feina cap risc

Cobreix-te
amb roba
llarga i ampla
per protegir
la pell del sol

Protegeix-te
el cap, la cara,
les orelles i el coll,
amb un barret
(d'ala ampla o amb
penjoll estil legionari).
Si utilitzes casc,
que estigui equipat amb
un penjoll d'estil legionari

Consulta
la radiació UV

Busca l'ombra.
No et quedis al sol,
sempre que sigui possible,
durant els períodes
amb més radiacions
ultraviolades (10 am - 3 pm).
Cal quedar-se a l'ombra
durant els descansos

**Fes ús de
protecció solar,**
SPF 30 o més.

**Utilitza
ulleres de sol**
amb protecció UV

Hidrata't

BAIX

UV INDEX 01

UV INDEX 02

UV INDEX 03

UV INDEX 04

UV INDEX 05

MODERAT

Necessita protecció: cal mantenir-se a l'ombra durant les hores centrals del dia i utilitzar roba de treball, protecció solar i barret.

UV INDEX 06

UV INDEX 07

ALT

UV INDEX 08

UV INDEX 09

UV INDEX 10

UV INDEX 11+

MOLT ALT

EXTREM

Necessita protecció extra: evita sortir durant les hores centrals del dia. Busca l'ombra. És imprescindible utilitzar roba de treball, protecció solar i barret.

Action plan to prevent the effects of heat waves on health (POCS)
Department of Health. Government of Catalonia

Infographic: In the face of the heat, we take prevention



Action plan to prevent the effects of heat waves on health (POCS)
Department of Health. Government of Catalonia

Annex 5. Risk groups and recommendations for professionals

Risk groups in the event of a heat wave

- Elderly people, especially the over 75s group.
- People with certain social circumstances (living alone, poverty, etc.) others).
- People with physical or mental disabilities who have limited self-care and mobility.
- Babies.
- People with chronic diseases: cardiorespiratory, diabetes, hypertension arterial, kidney diseases, Parkinson's, Alzheimer's, obesity, among others.
- People who receive diuretic medications or medications that act on the central nervous system.
- People with insufficient hydration or who engage in intense physical activity.
- Workers exposed to hot environments.

We will consider the **elderly** , those with physical or mental disabilities that limit their self-care, those who live alone and without anyone to take care of them and those with a history of risk (chronic diseases, medication that acts on the central nervous system, poverty, among others).

Recommendations for the management of the elderly and people at risk

a) Organizational considerations

- 1) Strengthening the control of fragile patients: Home care program (ATDOM).
- 2) Special monitoring in the summer of chronically ill people, those with dementia, those with mental health problems, polymedication, disabilities, among others.
- 3) Active intervention in fragile people. Promote self-care: hydration, control of general medications, system depressant medication central nervous system, ventilation, reduction of sun exposure and mobility, among others others.
- 4) Thermal adaptation in detention centers (social health centers, hospitals) acute, mental health).
- 5) Cooperation of health services (CAP, CSM, socio-health centers) with social services (homes or geriatric residences).
- 6) Recommendations for geriatric residences (annex 3).

b) Clinical and therapeutic considerations

Introduction

In summer, extreme temperatures can lead to several health problems. When, in a situation of extreme heat, there is a decrease in capacity of eliminating body heat, one enters a situation of risk of developing the heat-related illnesses. Certain circumstances increase the risk of suffering from these diseases: some people have a reduced ability to eliminate body heat and others suffer from certain underlying diseases that make increase their body temperature; these are, above all, the elderly, young children, patients in treatment with psychotropic drugs, alcoholics, drug addicts and the people with chronic illnesses. It is important to remember that the exhausting activities of summer carried out during the hours with the highest temperature also predispose to suffering some of the medical disorders associated with heat.

Definitions

Heat stress. Perception of discomfort and physiological stress associated with exposure to environmental heat, especially during physical work.

Heat stroke. Heat stroke occurs when a person produces a large amount of heat that cannot be eliminated through sweat or by dilating blood vessels. This occurs when very extreme temperatures are reached. It is a disease severe, characterized by a core temperature above 40°C and alterations in central nervous system, which can cause delirium, seizures or coma, such as consequence of exposure to environmental heat or extreme physical exercise. The skin It is hot, red and dry, without sweating.

Heat exhaustion. Moderately severe illness due to water loss or salt, due to exposure to high environmental heat or strenuous physical exercise. Signs and symptoms include intense thirst, fatigue, discomfort, anxiety, dizziness, weakness and headache; core temperature may be normal, below normal or slightly elevated (above 37°C but below 40°C).

Hyperthermia. Increase in body temperature above the regulatory level. hypothalamic when heat loss mechanisms are altered (by drugs or by disease) or overwhelmed by external (environmental or induced) or internal heat (metabolic).

Multiorgan dysfunction syndrome. Changes that occur in more than one organ after traumatic injuries, sepsis or heat stroke.

Clinical manifestations of extreme heat

Although the manifestations can be very diverse (fatigue, lipothymia, burns, edema, among others), there are two major diseases related to extreme heat: heat exhaustion and heat stroke.

Heat exhaustion manifests itself because the person feels very tired, has feeling dizzy and sweating profusely. The skin is cool and moist, the pulse is rapid and the breathing is shallow and rapid. In this case, it is advisable to drink fresh water, shower and stay in air-conditioned places. If symptoms do not go away and does not recover in about thirty minutes, if an alteration in the level of conscience or if the person has an underlying illness, it is advisable to request one urgent medical assessment.

Heat stroke is a serious clinical condition, in which the mechanism that allows the sweating begins to fail and the body can no longer cool itself. It manifests with very high hyperthermia, usually above 40 °C, alteration of the level of consciousness (which can range from a character disorder to a state of coma) and lack of sweating. The skin is red, hot and dry. Requires a

urgent medical attention, as it can lead to the patient's death if not treated promptly
immediate medical treatment.

Definition of a suspected case of heatstroke

- Continuous exposure to high temperatures and humidity.
- Body temperature $\geq 40^{\circ}\text{C}$ and other causes have been reasonably excluded causes of hyperthermia (cocaine intake, among others).
- Finding the person dead, with no evidence of any other cause, during the wave of heat.

Clinical manifestations of heat stroke:

- Hyperthermia (body temperature $\geq 40^{\circ}\text{C}$).
- Altered mental status (usually severe, but can vary from dizziness, disorientation, even lethargy, convulsions, delirium and coma).
- Anhidrosis (dry, red, hot skin without sweating).
- Tachycardia and hyperventilation.
- There may be hypotension.

Complications of heatstroke:

- **Multiorgan dysfunction** syndrome : encephalopathy, renal failure acute, rhabdomyolysis, acute respiratory distress syndrome, myocardial failure, liver failure, intestinal ischemia or infarction, pancreatic failure, complications bleeding, especially disseminated intravenous coagulation.

c) Treatment

The **best measure** for treating heat-related illnesses is **prevention**.

It is advisable to drink plenty of fluids (alcoholic or sugary drinks are harmful, because they facilitate the loss of fluids). Sports drinks provide many ions and allow the replacement of mineral salts lost through sweat. It is

It is important to drink very regularly, anticipating the feeling of thirst. Drinks very fresh are not advisable, as they make the feeling of thirst disappear more quickly. feeling of thirst and, therefore, less fluids are ingested.

People who have underlying diseases that require them to limit their intake of liquids are advised to consult their doctor to find out to what extent amount they can drink during the summer.

It is important to wear appropriate clothing: light and light-colored. You should try to be in places with air conditioning. If a person does not have one at home, it is advisable to visit public places that have them, such as cinemas, libraries or large shopping centers. It is advisable not to leave anyone in a car parked, with the windows closed. People who have to work exposed to the heat it is important that they remember the need to drink non-carbonated drinks very often and, if they have to have strenuous physical activities, they must rest in the shade periodically. Those who have elderly relatives who live alone, They are advised to check them often to see if they have any of the following: symptoms related to heatstroke. If we find a person who presents any of these symptoms, you should take him to the shade or to a place with air conditioned, and try to lower his body temperature by any method (by immersing it in cold water, for example) **while seeking medical attention urgent**.

Once in the hospital, rehydration measures will be initiated and measures to lower the body temperature.

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Recommendations for hydration for the elderly

Calculate your daily fluid intake goal: for the first 20 kg of weight = 1,500 ml; additional 15 ml/kg from 20 kg of weight. Water should represent 70-75% of the calculated intake.

Compare the ideal intake with the actual one to find out the deficit.

Ensure the necessary fluid intake, considering that the administration of a drug via

Oral administration must be accompanied by 180 ml of water.

Fluids should be administered from 8 a.m. to 7 p.m. No

It is advisable to take them after this time due to the risk of urinary incontinence.

night and sleep fragmentation.

Drinking too much pure water can lead to severe hyponatremia which can lead to complications and death. The addition of sodium chloride and other soluble substances to the drink (20-50 mmol/l) reduces water loss through urine and facilitates recovery of electrolyte balance.

Situations with risk of dehydration

- Dementia, depression, stroke, diabetes mellitus, infections, malnutrition, urinary incontinence, history of dehydration.
- Drugs: diuretics, neuroleptics, antidepressants, anxiolytics, laxatives, corticosteroids.
- Age: > 75 years and infants.
- Functional dependence for activities of daily living.
- Inadequate nutritional status, including high protein intake.
- Acute situations with vomiting, diarrhea, fever.
- Consumers of illegal drugs and alcohol.

Annex 6. Heatstroke notification form

-Date of declaration: _____ UVE _____

-Declaring Center _____ Admission date _____ Discharge date _____

-Patient population: _____ Population where the heatstroke is suffered: _____

-Name: _____

-Age: _____ -Gender: Home Dona

-Exposure to high temperatures: _____ Yeah _____ no

Specify the environmental risk factor (1): _____

-Background of individual risk factors (2)

Basic diseases: _____ Yeah _____ no

Please specify: _____

Previous treatments: _____ Yeah _____ no

Specify them: _____

-Symptoms: Date of onset ____ / ____ / ____

-Alteration of mental status:

-High body temperature Please specify. _____

-Anhidrosis

-Tachycardia Please specify. _____

-Hypotension Please specify. _____

-Disseminated intravenous coagulation Rhabdomyolysis

-Multiorgan dysfunction

-Dehydration

-Hyponatremia

-Others: _____

-Admission to ICU: Yes _____ No _____

-Have other causes of hyperthermia been ruled out? Yes _____ No _____

Specify: (cocaine intake, etc.). _____

-Evolution: _____

-Diagnostic guidance _____

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¹ **Exposure to environmental risk factors:**

*Presence of an **environmental or social risk** factor (for example: people living alone, on the street and/or in unfavorable conditions, lack of air conditioning and housing that is difficult to refrigeration), excessive exposure to heat for **work** reasons (manual work outdoors or requires high contact with hot environments), excessive exposure to heat for reasons **sports** (sports of high physical intensity) or **leisure** (continuous exposure for different days at high temperatures that are maintained at night).*

² **Individual risk factors:** cardiovascular, respiratory and mental diseases (dementia, Parkinson's, Alzheimer's...); chronic diseases (diabetes mellitus, obesity morbid...); medical treatments (diuretics, neuroleptics, anticholinergics and tranquilizers); memory disorders, difficulties with understanding or orientation or little autonomy in life daily; Acute illness during an episode of excessive temperature; alcohol consumption and other drugs.

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Annex 7. Heatstroke death notification form (IMLCFC)

-Date of declaration: _____ Date of autopsy _____

-Declaring Center _____ Date of admission of the body to the SPF _____

-Population of the deceased: _____ Population where he suffered from heatstroke: _____

-Name: _____

-Age: _____ -Gender: Home Dona

-Exposure to high temperatures: _____ Yeah _____ no

Specify the environmental risk factor (1): _____

-Background of individual risk factors (2)

Basic diseases: _____ Yeah _____ no

Please specify: _____

Previous treatments: _____ Yeah _____ no

Specify them: _____

-Admission to ICU: Yes _____ No _____

-Have other causes of hyperthermia been ruled out? Yes _____ No _____

Specify them: (cocaine intake, etc.). _____

-Evolution: _____

-Diagnostic guidance _____

Exposure to an environmental risk factor: Presence of an **environmental or social risk** factor (for example example: people who live alone, on the street and/or in unfavorable conditions, absence of air conditioning and homes that are difficult to cool), excessive exposure to heat for **work** reasons (manual work outdoors or requiring high contact with hot environments), **sports** (sports of high physical intensity) or **leisure** (continuous exposure over several days to high temperatures that are maintained at night).

² **Individual risk factors** cardiovascular, respiratory and mental diseases (dementia, Parkinson's, Alzheimer's...); chronic diseases (diabetes mellitus, morbid obesity...); medical treatments (diuretics, neuroleptics, anticholinergics, and tranquilizers); memory disorders, difficulties with understanding or orientation or little autonomy in daily life; acute illness during an episode excessive temperature; consumption of alcohol and other drugs.

SPF: Professional funeral services